Min Jae Bae, Ph.D.

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Education

- Ph.D. in Biomedical Science and Engineering, Konkuk University, 2021 – Present (expected Feb. 2027) — Advisor: Prof. Dae-Jin Yun
- M.S. in Biomedical Science and Engineering, Konkuk University, 2019 – 2021 — Thesis: Structural and Functional Analysis of HOS15, a Substrate Receptor of CRL4 in Cold Stress Signaling
- B.S. in Biomedical Science and Engineering, Konkuk University, 2013 – 2019 — Thesis: Functional Studies of SUMOylation of Histone Deacetylase in Plants

Research Focus

My research investigates the epigenetic regulation of plant stress adaptation, focusing on the linker histone H1 and chromatin remodeling in *Arabidopsis thaliana*. Building on earlier work on HOS15–PWR–HDAC complexes in cold stress signaling, I study how histone H1 controls chromatin accessibility and transcriptional reprogramming under abiotic stress, using CRISPR-based genetics, ChIP-seq, RNA-seq, and confocal imaging.

Research Projects

- Global Plant Stress Research Center (NRF), 2024.08–2030.04 — Participating Researcher
- Genome-wide chromatin reprogramming and transcriptome regulation in plant environmental stress adaptation (NRF), 2026.03–2031.02 — Participating Researcher
- Chromatin reconfiguration and transcriptional control for plant stress adaptation (NRF), 2022.03–2027.02 — Co-participating Researcher

Professional Service

- Member, NASA GeneLab OSDR Plant Analysis Working Group (Plant AWG), Sep. 2026–Present — Contributing to the analysis of spaceflight and astrobotany omics datasets within NASA's Open Science Data Repository (OSDR), collaborating with international researchers on plant biology under the space environment.
- Member, NASA GeneLab OSDR Regolith Analysis Working Group (Regolith AWG), Sep. 2026–Present — Participating in cross-disciplinary efforts on plant and plant–microbe growth in lunar/Martian regolith, contributing plant molecular biology expertise to regolith-related dataset curation and standards within NASA OSDR.

Skills

- **Plant molecular biology:** Gateway cloning, Gibson assembly, vector construction and design (Vector NTI), transgenic line generation, Arabidopsis genetics and phenotypic analysis
- **Epigenetics & chromatin biology:** histone biology, chromatin remodeling, nucleosome organization, ChIP-seq, open-chromatin/accessibility analysis
- **Omics & data analysis:** RNA-seq and NicE-seq analysis using SalmonProcessor and OmicsForge, LC-MS proteomics, alternative splicing analysis, sequence alignment (BioEdit), whole-genome data visualization with IGV
- **Protein & stress biology:** ubiquitin–proteasome pathway, E3 ligase complexes, protein stability, cold/drought/thermotolerance and senescence signaling
- **Software/tools:** SalmonProcessor, OmicsForge, IGV, BioEdit, Vector NTI, HOMER, GraphPad Prism, Bio-Rad CFX Maestro (RT-qPCR analysis), confocal imaging analysis (Zeiss ZEN, Leica LAS X), ImageJ

Publications (SCIE/ESCI)

1. Ali A, Zareen S, Bader ZE, Park J, Khan IU, Park K, Albakri N, **Bae MJ**, Bressan RA, Pardo JM, Yun DJ, Xu ZY. "A GPS2-like protein interacts with HOS15 and HDA6 to form a repressor complex that regulates ABA signaling and drought adaptation in Arabidopsis." *Plant Communications*, 2026.
2. **Bae MJ**, Zareen S, Yun H, Ali A, Bader ZE, Park K, Albakri N, Jung YJ, Yun DJ, Park J. "Unveiling the epigenetic and stress-responsive function of histone H1 in plants." *Plant Signaling & Behavior*, 2026.
3. Albakri N, Bader ZE, Park KS, **Bae MJ**, Zareen S, Yun H, Pradhan S, Yun DJ, Park J. "HOS15 Modulates Alternative Splicing Homeostasis Under Cold Stress in Arabidopsis Thaliana." *Journal of Plant Biology*, 2025.
4. Park K, Lim CJ, Albakri N, **Bae MJ**, Park J. "HOS15 Contributes to Thermotolerance Through Destabilization of HD2C in Arabidopsis." *Journal of Plant Biology*, 2025.
5. Zareen S, Ali A, **Bae MJ**, Albakri N, Park K, Yun H, Yun DJ, Park J. "HOS15 together with PWR–HDA9 positively regulates dark-induced senescence in Arabidopsis." *Plant Signaling & Behavior*, 2025.
6. Bader ZE, **Bae MJ**, Ali A, Park J, Baek D, Yun DJ. "GIGANTEA-ENHANCED EM LEVEL complex initiates drought escape response via dual function of ABA synthesis and flowering promotion." *Plant Signaling & Behavior*, 2023.
7. Lim CJ, Park KS, Ali A, Park J, Ryou SM, Shen M, Khan HA, Bader ZE, Zareen S, **Bae MJ**, Choi JH, Xu ZY, Pardo JM, Yun DJ. "Negative regulation of floral transition in Arabidopsis by HOS15-PWR-HDA9 complex." *Frontiers in Plant Science*, 2023.
8. Lim CJ, Park J, Shen M, Park HJ, Cheong MS, Park K, Baek D, **Bae MJ**, Ali A, Jan M, Lee SY, Lee BH, Kim WY, Pardo JM, Yun DJ. "The Histone-Modifying Complex PWR/HOS15/HD2C Epigenetically Regulates Cold Tolerance." *Plant Physiology*, 2020.

Manuscripts in preparation (targeting PNAS): Mechanisms of Histone H1 degradation under cold stress; HOS15-dimerization promotes HD2C degradation during Cold Stress.

Patents

- Vector for improving productivity of cellulose and transformed mouseear cress by the same — KR 10-2023-0113597
- Vector for improving productivity and storage capacity of vegetable oil and transformed plant by the same — KR 10-2024-0168464
- Vector for improving productivity of cellulose and transformed mouseear cress by the same — KR 10-2024-0168432
- Sucrose-accumulating plant and method for preparing the same — KR 10-2025-0104961
- Ethanol production and tolerance enhanced yeast strain and method for producing ethanol using the same — KR 10-2025-0104960
- Recombinant Kluyveromyces marxianus strain with engineered xylose metabolic pathway and method for producing ethanol using the same — KR 10-2025-0104960

Awards and Honors

- HOPE Fellow, HOPE Meetings with Nobel Laureates, Japan Society for the Promotion of Science (JSPS), Japan, Mar. 2024
- Excellent Poster Award, Korean Society of Plant Biologists (KSPB), Korea, Nov. 2021

Conference Presentations

- 2026 GPSRC International Conference, Konkuk University, Korea, May 2026
- 2026 Winter Conference of the Korean Society of Plant Biologists, Korea University, Korea, Feb. 2026 — Poster
- 2nd Global Plant Stress Research Center (GPSRC) & 2025 KSMCB Plant Molecular Biology International Symposium, Seoul, Korea, Aug. 2025 — Organizer
- 15th HOPE Meeting, Kyoto, Japan, Feb. 2024 — Poster
- 2024 International Conference of the Korean Society of Plant Biologists, Seogwipo-si, Korea, Nov. 2024 — Oral Presentation
- 2022 International Conference of the Korean Society of Plant Biologists, Busan, Korea, Nov. 2022 — Poster
- 2021 International Conference of the Korean Society of Plant Biologists, Online, Nov. 2021 — Poster
- 2020 International Conference of the Korean Society of Plant Biologists, Online, Nov. 2020 — Poster
- 2019 Annual Conference of the Korean Society of Plant Biologists, Buyeo, Korea, Nov. 2019 — Poster
- 2018 International Conference of the Korean Society of Plant Biologists, Busan, Korea, Nov. 2018 — Poster